

GRÖBNER BASES AND CONVEX POLYTOPES

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1. LESSON 1: GRÖBNER BASES

1.1. **Introduction.** We follow

- ‘Gröbner bases and convex polytopes’ by Sturmfels.
- ‘Ideals, Varieties, and Algorithms’ by Cox, Little, and O’Shea.

1.2. **Definitions and notation.** Let k be a field (I think most of this work in general, but let’s assume algebraically closed just in case). We write the polynomial ring over k in n variables as

$$k[x_1, \dots, x_n]$$

Definition 1.1. The ideal I generated by $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ is defined as

$$I = (f_1, \dots, f_m) = \{h_1 f_1 + \dots + h_m f_m \mid h_i \in k[x_1, \dots, x_n]\}$$

Exercise 1.1. Show that $I = (f_1, \dots, f_m)$ really is an ideal.

Note that $I = (f_1, \dots, f_m)$ is a *finitely generated* ideal. We call $\{f_1, \dots, f_m\}$ a basis of I . If $I = (f)$ then we call it *principal*.

In general ideals (when I am thinking about any rings, not just polynomial rings), do not have to be finitely generated. It turns out that in the polynomial ring case, every ideal is finitely generated, which is very nice. This is Hilbert’s Basis Theorem, which we will state and prove later

Remark 1.1. Why do people care so much about ideals in polynomial rings? This is because, if you are given an ideal $I = (f_1, \dots, f_m) \subset k[x_1, \dots, x_n]$, then we can consider a geometry object (a *variety*) defined as

$$V(I) = \{(a_1, \dots, a_n) \in \mathbb{R}^n \mid f_1(a_1, \dots, a_n) = \dots = f_m(a_1, \dots, a_n) = 0\}$$

i.e. the zero set of the polynomials. So, we can think that if we want to study geometric properties of $V(I)$ we want to write I in a nice way, hence we need to study how to write a nice generating set of the ideal.

Note that you can go back as well

$$I(V) = \{f \in k[x_1, \dots, x_n] \mid f(a_1, \dots, a_n) = 0 \text{ for all } (a_1, \dots, a_n) \in V\}$$

Unfortunately, $I(V(I)) \neq I$, but at least we have $I \subset I(V(I))$. To see this look at $I = (x^2) \subset k[x]$. We have

$$I(V(I)) = I(\{x = 0\}) = (x)$$

A reasonable question is: since every ideal in a polynomial ring is finitely generated, can we build a basis? This is the question that **Gröbner bases** (a fundamental building block in computational algebraic geometry) answers.

1.3. **Polynomials in $k[x]$.** Let’s think about polynomials in one variable for a second. Life is easy.

Definition 1.2. Let $f \in k[x]$ be a non-zero polynomial. Then we can write

$$f = a_0 + a_1 x + \dots + a_m x^m$$

We say $m = \deg(f)$ (the degree of the polynomial) and $a_m = \text{LT}(f)$ (the leading term of the polynomial).

Proposition 1.1. Let $g \in k[x]$ be a non-zero polynomial. Then every $f \in k[x]$ can be written uniquely in the form

$$f = gq + r$$

where $q, r \in k[x]$ with $\deg(r) < \deg(g)$ or $r = 0$.

Proof. Let $q = 0$, set $r = f$. If $\deg(f) < \deg(g)$ or $r = 0$ we are done. Otherwise set

$$q' = q + \frac{\text{LT}(r)}{\text{LT}(g)} \quad r' = r - \frac{\text{LT}(r)}{\text{LT}(g)} g$$

Then clearly $f = q'g + r'$. This process (inductively) reduces the degree of r , so we have shown existence.

For the uniqueness, suppose

$$f = qg + r = q'g + r'$$

with $\deg(r) < \deg(g)$ or $r = 0$ and $\deg(r') < \deg(g)$ or $r' = 0$. If $r \neq r'$, then $\deg(r - r') < \deg(g)$. This also implies $q \neq q'$, leading to a contradiction

$$\deg(r - r') = \deg((g - g')q) = \deg(q - q') + \deg(g) \geq \deg(g)$$

□

Corollary 1.1. *Every ideal I in $k[x]$ is principal. Furthermore, if $I = (f)$ for some $f \in k[x]$ then f is unique up to multiplication by a non-zero scalar in k .*

Proof. If $I = (0)$ we are done.

Otherwise, let $f \in I$ be a non-zero polynomial of minimum degree. We claim $(f) = I$. Let $g \in I$. Then $g = qf + r$ for $r = 0$ or $\deg(r) < \deg(f)$. However, $r \in I$, so by minimality of degree of f we must have $r = 0$. So $I = (f)$.

Assume $(f) = (g)$, then $f = hg$ and $g = kf$ for some $h, k \in k[x]$. Therefore $\deg(g) = \deg(f)$ so they differ by multiplication by a non-zero constant. □

1.4. Polynomials in $k[x_1, \dots, x_n]$. The aim of this section is to be able to generalise the results from the previous section to an arbitrary polynomial ring $k[x_1, \dots, x_n]$. We will see that the nice picture from $k[x]$ is not fully recovered.

First we need a way to order monomials in multiple variables. Recall that monomials are things like $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ for $(\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$. We have a 1-1 correspondence between monomials in $k[x_1, \dots, x_n]$ and $\mathbb{Z}_{\geq 0}^n$.

Definition 1.3. A **total order** on $\mathbb{Z}_{\geq 0}^n$ is defined as a binary relation $>$ such that

- For all $\alpha, \beta, \gamma \in \mathbb{Z}_{\geq 0}^n$, $\alpha > \beta$ and $\beta > \gamma$ implies $\alpha > \gamma$;
- For all $\alpha, \beta \in \mathbb{Z}_{\geq 0}^n$ we have one of the following: $\alpha > \beta$ or $\alpha = \beta$ or $\beta > \alpha$.

We can use a total order on $\mathbb{Z}_{\geq 0}^n$ to define a *monomial order* on monomials in $k[x_1, \dots, x_n]$.

Definition 1.4. Let $>$ be a total order on $\mathbb{Z}_{\geq 0}^n$ such that

- If $\alpha > \beta$ then $\alpha + \gamma > \beta + \gamma$ for all $\gamma \in \mathbb{Z}_{\geq 0}^n$,
- For any non-empty subset of $\mathbb{Z}_{\geq 0}^n$ there exists a smallest element with respect to $>$

Then a **monomial order** is a binary relation $\succ$ on monomials in $k[x_1, \dots, x_n]$ such that $\mathbf{x}^\alpha \succ \mathbf{x}^\beta$ if and only if $\alpha > \beta$.

Example 1.1.

- (Lexicographic order) $\alpha >_{\text{lex}} \beta$ if the left-most non-zero entry of $\alpha - \beta$ is positive. For example if $x_1 > x_2 > x_3$, then $x_1^2 x_2^3 \succ_{\text{lex}} x_1^2 x_2^2 x_3^5$ because the left-most non-zero entry of $(2, 3, 0) - (2, 2, 5)$ is positive.
- (Graded lexicographic order) $\alpha >_{\text{grlex}} \beta$ if $\sum_i \alpha_i > \sum_i \beta_i$ or $\sum_i \alpha_i = \sum_i \beta_i$ and $\alpha >_{\text{lex}} \beta$. Now $x_1^2 x_2^2 x_3^5 \succ_{\text{grlex}} x_1^2 x_2^3$ because $\sum_i \alpha_i = 5 < 9 = \sum_i \beta_i$.

Definition 1.5. Let $f \in k[x_1, \dots, x_n]$ and $\succ$ be a monomial order. We can write $LT(f)$ as the leading term of f with respect to $\succ$, i.e. the term of f with the largest monomial with respect to $\succ$.

Theorem 1.1. *Let $\succ$ be a monomial order on $k[x_1, \dots, x_n]$ and $f_1, \dots, f_s \in k[x_1, \dots, x_n]$. Then every $f \in k[x_1, \dots, x_n]$ can be written in the form*

$$f = h_1 f_1 + \cdots + h_s f_s + r$$

where $h_i, r \in k[x_1, \dots, x_n]$ and no term of r is divisible by any of the leading terms $LT(f_i)$.

This is almost like the division algorithm in $k[x]$, but we have a problem: the representation of f is not unique. For example, let $f_1 = x_1$ and $f_2 = x_1 + x_2$. Then we can write x_1 as

$$x_1 = 1 \cdot f_1 + 0 \cdot f_2 + 0$$

or as

$$x_1 = 2 \cdot f_1 - 1 \cdot f_2 + x_2$$

We won't be able to recover from the non-uniqueness of the h_i 's, but we are able to do something about the non-uniqueness of r . This is what Gröbner bases are for.

1.5. Gröbner bases.

Definition 1.6. Let I be an ideal in $k[x_1, \dots, x_n]$. If

$$I = (x_\alpha \mid \alpha \in A)$$

for some $A \subset \mathbb{Z}_{\geq 0}^n$, then we call I a **monomial ideal**.

Theorem 1.2 (Dickson's Lemma). *Every monomial ideal in $k[x_1, \dots, x_n]$ is finitely generated.*

Definition 1.7. Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ a monomial order. The **initial ideal** of I with respect to $\succ$ is the monomial ideal generated by the leading terms of the elements of I , i.e.

$$\text{in}_\succ(I) = (LT(f) \mid f \in I)$$

Example 1.2. Unfortunately, the initial ideal of an ideal I is not necessarily finitely generated by the leading terms of a finite generating set of I . For example, let $I = (x_1, x_1 + x_2)$ with $x_1 \succ x_2$. Then $(LT(x_1), LT(x_1 + x_2)) = (x_1)$, which is strictly contained in $\text{in}_\succ(I)$ since it does not contain x_2 .

Theorem 1.3 (Hilbert's Basis Theorem). *Every ideal in $k[x_1, \dots, x_n]$ is finitely generated.*

Proof. Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ a monomial order. Then $\text{in}_\succ(I)$ is a monomial ideal, so by Dickson's Lemma it is finitely generated, so there exist $f_1, \dots, f_m \in I$ such that $\text{in}_\succ(I) = (LT(f_1), \dots, LT(f_m))$. We claim that $I = (f_1, \dots, f_m)$.

Let $f \in I$. We can write

$$f = h_1 f_1 + \dots + h_m f_m + r$$

where no term of r is divisible by any of the leading terms $LT(f_i)$. If $r = 0$ we are done. Otherwise, since $f, h_i, f_i \in I$, we have that $r = f - (h_1 f_1 + \dots + h_m f_m) \in I$. This implies that $LT(r) \in \text{in}_\succ(I)$, so there exists some i such that $LT(f_i)$ divides $LT(r)$, which is a contradiction. $\square$

So, not only we have shown that every ideal in $k[x_1, \dots, x_n]$ is finitely generated, but we have also shown that we can find a finite generating set of I such that the leading terms of the generators generate the initial ideal of I . This motivates the following definition.

Definition 1.8. Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ a monomial order. A finite subset $G = \{g_1, \dots, g_t\}$ of I is called a **Gröbner basis** of I with respect to $\succ$ if the ideal generated by the leading terms of the elements of G is equal to the ideal generated by the leading terms of the elements of I , i.e.

$$(LT(g_1), \dots, LT(g_t)) = (LT(f) \mid f \in I)$$

So, in the theorem above we have shown that every ideal in $k[x_1, \dots, x_n]$ has a Gröbner basis and that a Gröbner basis is a generating set of the ideal. However, a Gröbner basis does not have to be unique, and it might have many redundancies.

Gröbner bases are very useful because they allow us to recover the uniqueness of r that we had lost in our new division algorithm.

Proposition 1.2. *Let $f \in k[x_1, \dots, x_n]$. Let $I = (g_1, \dots, g_t)$ be a Gröbner basis of I with respect to $\succ$. Then, $f \in I$ if and only if the remainder on division of f by $g_1, \dots, g_t$ is zero.*

The h_i 's are still not unique, but this recovers a lot of the unhappiness we were feeling before in our division algorithm.

1.6. Computing Gröbner bases. Computing Gröbner bases is a very important problem in computational algebraic geometry. The most famous algorithm to compute Gröbner bases is Buchberger's algorithm, which underpins a lot of different computer algebra computations. We won't go into the details of the algorithm, but we will just state it and give some intuition on how it works.

Definition 1.9. Let $f, g \in k[x_1, \dots, x_n]$ and $\succ$ a monomial order. The **S-polynomial** of f and g is defined as

$$S(f, g) = \frac{\text{lcm}(LT(f), LT(g))}{LT(f)} f - \frac{\text{lcm}(LT(f), LT(g))}{LT(g)} g$$

The S-polynomial is a way to measure how much f and g fail to be a Gröbner basis.

Theorem 1.4 (Buchberger's Criterion). *Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ a monomial order. A finite subset $G = \{g_1, \dots, g_t\}$ of I is a Gröbner basis of I with respect to $\succ$ if and only if the remainder on division of $S(g_i, g_j)$ by G is zero for all $i \neq j$.*

Buchberger's algorithm is an iterative algorithm that starts with a generating set of I and then computes the S-polynomials of the generators. If the remainder on division of any S-polynomial is not zero, then we add it to our generating set and repeat the process until we get a Gröbner basis. This process terminates since we create a strictly increasing sequence of monomial ideals, and there are no infinite strictly increasing sequences of monomial ideals (this is a consequence of Dickson's Lemma).

Definition 1.10. A Gröbner basis G of an ideal I is called **reduced** if for all $g \in G$ we have that the coefficient of $LT(g)$ is 1 and no term of g is divisible by any of the leading terms of the other elements of G .

Theorem 1.5. *Every ideal in $k[x_1, \dots, x_n]$ has a unique reduced Gröbner basis with respect to a given monomial order.*

Note that we can also define a more universal object, called a *universal Gröbner basis*, which is a finite subset of I that is a Gröbner basis of I with respect to every monomial order. However, we won't go into the details of this object, but it is worth mentioning that it exists (the fact that it exists is a consequence of the fact that there are only finitely many initial ideals of I – this will become obvious when we introduce the concept of Gröbner fan and state polytope).

2. LESSON 2: FROM ALGEBRA TO COMBINATORICS

2.1. Some combinatorics.

Definition 2.1. A **polyhedron** is a subset of $\mathbb{R}^n$ defined by a finite intersection of half spaces in $\mathbb{R}^n$, i.e.

$$P = \{x \in \mathbb{R}^n \mid Ax = b\}$$

where $b \in \mathbb{R}^m$ and $A \in \mathbb{R}^{m \times n}$. If $b = 0$, then P is a polyhedral cone.

Definition 2.2. A **polytope** is a bounded polyhedron. Equivalently, a polytope is the convex hull of a finite set of points in $\mathbb{R}^n$.

Definition 2.3. Let $P \subset \mathbb{R}^n$ be a polytope. A **face** of P is a subset of P defined by

$$F_w(P) = \{x \in P \mid w \cdot x \geq w \cdot y \text{ for all } y \in P\}$$

for some $w \in \mathbb{R}^n$.

Example 2.1. Consider the square $P = \text{conv}\{(0,0), (1,0), (1,1), (0,1)\} \subset \mathbb{R}^2$. Taking $w = (1,0)$, we maximise the dot product $w \cdot x = x_1$ over P , giving

$$F_{(1,0)}(P) = \text{conv}\{(1,0), (1,1)\},$$

which is the right edge of the square. Taking $w = (1,1)$, we maximise $x_1 + x_2$, giving the vertex $F_{(1,1)}(P) = \{(1,1)\}$. Taking $w = (0,0)$, we get $F_{(0,0)}(P) = P$ itself, so P is a face of itself. The faces of the square are: the polytope P itself, its four edges, its four vertices, and the empty face (obtained by taking w in the interior of the outer normal cone of no face, though one often defines the empty set as a face by convention).

Definition 2.4. A **fan** is a collection of polyhedral cones in $\mathbb{R}^n$ such that

- If σ is a cone in the fan, then every face of σ is also a cone in the fan.
- If σ_1 and σ_2 are cones in the fan, then their intersection $\sigma_1 \cap \sigma_2$ is a face of both σ_1 and σ_2 .

Definition 2.5. Let $P \subset \mathbb{R}^n$ be a polytope. Let F be a face of P . We define the **normal cone** of F as

$$\mathcal{N}_F = \{w \in \mathbb{R}^n \mid F_w(P) = F\}$$

The **normal fan** of P is the collection of normal cones of all faces of P .

Example 2.2. Consider the triangle $P = \text{conv}\{(0,0), (1,0), (0,1)\} \subset \mathbb{R}^2$. The vertex $(1,0)$ is the unique maximiser of $w \cdot x$ whenever $w = (w_1, w_2)$ satisfies $w_1 > 0$ and $w_1 > w_2$. Hence the normal cone of the vertex $(1,0)$ is the open cone $\{(w_1, w_2) \mid w_1 > 0, w_1 > w_2\}$ (together with appropriate boundary rays to form its closure). Similarly, the normal cone of the edge $\text{conv}\{(1,0), (0,1)\}$ is the ray $\{t(1,1) \mid t \geq 0\}$, since $w = (1,1)$ gives $w \cdot (1,0) = w \cdot (0,1) = 1 > 0 = w \cdot (0,0)$, so both vertices of that edge are simultaneously maximised. The normal cones of all faces of P partition $\mathbb{R}^2$ into a complete fan with three maximal cones (one per vertex), three rays (one per edge), and the origin (corresponding to P itself).

2.2. Newton polytopes. For a given multivariate polynomial, we can associate a polytope to it, called the **Newton polytope**. Let's look at how to construct this and how to generalise this to an ideal rather than just a unique polynomial.

Definition 2.6. Let $f \in k[x_1, \dots, x_n]$ be a polynomial. The **Newton polytope** of f is the convex hull of the exponent vectors of the monomials appearing in f .

Example 2.3. Let $f = 3x^2y + xy^2 + 2x^3 + y^3 \in k[x, y]$. The exponent vectors of the monomials appearing in f are $(2,1)$, $(1,2)$, $(3,0)$, and $(0,3)$. Thus

$$\text{Newt}(f) = \text{conv}\{(2,1), (1,2), (3,0), (0,3)\}.$$

This is a quadrilateral in $\mathbb{R}^2$. Note that the Newton polytope depends only on the *support* of f (which monomials appear), not on the values of the nonzero coefficients.

Definition 2.7. Let P, Q be polytopes in $\mathbb{R}^n$. The **Minkowski sum** of P and Q is defined as

$$P + Q = \{p + q \mid p \in P \text{ and } q \in Q\}$$

Example 2.4. Let $P = \text{conv}\{(0,0), (1,0)\}$ be a horizontal unit segment and $Q = \text{conv}\{(0,0), (0,1)\}$ a vertical unit segment. Then

$$P + Q = \text{conv}\{(0,0), (1,0), (0,1), (1,1)\},$$

which is the unit square. Geometrically, the Minkowski sum “sweeps” one polytope along the other: sliding the vertical segment along the horizontal one traces out the square.

Proposition 2.1. Let $f, g \in k[x_1, \dots, x_n]$ be polynomials. Then,

$$\text{Newt}(f \cdot g) = \text{Newt}(f) + \text{Newt}(g)$$

Proof. Let $f = \sum_{\alpha \in A} c_\alpha x^\alpha$ and $g = \sum_{\beta \in B} d_\beta x^\beta$ (where $c_\alpha \neq 0$ for all $\alpha \in A$ and $d_\beta \neq 0$ for all $\beta \in B$).

Showing $\text{Newt}(f \cdot g) \subset \text{Newt}(f) + \text{Newt}(g)$ is trivial since every monomial in fg will have exponent $\gamma = \alpha + \beta$ for $\alpha \in A$ and $\beta \in B$.

For the other direction, we note that a vertex of $\text{Newt}(f) + \text{Newt}(g)$ comes from $v + w$ where v is a vertex of $\text{Newt}(f)$ and w is a vertex of $\text{Newt}(g)$. It suffices to show that every such vertex $v + w$ is in $\text{Newt}(f \cdot g)$, i.e. that the monomial x^{v+w} actually appears in fg with nonzero coefficient. Since $v + w$ is a vertex of the Minkowski sum, there exists a weight vector $u \in \mathbb{R}^n$ such that v is the unique maximiser of $u \cdot \alpha$ over $\alpha \in A$, and w is the unique maximiser of $u \cdot \beta$ over $\beta \in B$. The coefficient of x^{v+w} in fg is

$$\sum_{\substack{\alpha \in A, \beta \in B \\ \alpha + \beta = v + w}} c_\alpha d_\beta.$$

If $\alpha + \beta = v + w$ with $\alpha \in A$ and $\beta \in B$, then $u \cdot \alpha + u \cdot \beta = u \cdot v + u \cdot w$. Since $u \cdot \alpha \leq u \cdot v$ and $u \cdot \beta \leq u \cdot w$, equality forces $\alpha = v$ and $\beta = w$ by uniqueness. Hence the sum above reduces to the single term $c_v d_w \neq 0$, confirming that x^{v+w} appears in fg . Since the vertices of $\text{Newt}(f) + \text{Newt}(g)$ all lie in $\text{Newt}(f \cdot g)$, and $\text{Newt}(f \cdot g)$ is convex, we obtain $\text{Newt}(f) + \text{Newt}(g) \subseteq \text{Newt}(f \cdot g)$. $\square$

We note that the Newton polytope's vertices correspond to each monomials of the polynomial. In some way, we would like to extend this to a combinatorial construction for an ideal, where each vertices of a polytope correspond to different initial ideals under different monomial orders. This is where the motivation for defining the *state polytope* (which will be the focus of the rest of the lecture) is from.

Example 2.5. The key point here is that for a principal ideal the generator is unique up to multiplication by a non-zero scalar. This is no longer true when we look at non-principal ideals, so looking at properties of specific generators stops making a lot of sense. For example, a naive way of trying to get some sort of Newton polytope for ideals would be to look at the Newton polytopes of the generators. Take for example

$$I = (x^2 - yz, xy - z^2) = (x^2 - yz, x^2 + xy - yz - z^2)$$

For the first set of generators the newton polytopes would be two segments, whilst looking at the second set of generators one of the segments has become a quadrilateral. This is not very meaningful, we cannot be looking at specific generators, instead we have to look at the possible initial ideals of the ideal.

2.3. Initial ideals revisited. So we have seen that as we vary the monomial order we get different Gröbner bases, and we have seen that the leading terms of the elements of a Gröbner basis generate the initial ideal of I . Understanding how the initial ideal of I changes as we change the monomial order is a very interesting problem, and it turns out that it has a very nice combinatorial structure.

First of all, we need to look at monomial orders from a more combinatorial point of view.

Definition 2.8. Let $w \in \mathbb{R}^n$. Let $f = \sum_{\alpha \in A} c_{\alpha} x^{\alpha}$ be a polynomial in $k[x_1, \dots, x_n]$. We define the **weight** of f with respect to w as

$$\text{wt}_w(f) = \max_{c_{\alpha} \neq 0} \{w \cdot \alpha\}.$$

and we define the **initial form** of f with respect to w as

$$\text{in}_w(f) = \sum_{w \cdot \alpha = \text{wt}_w(f)} c_{\alpha} x^{\alpha}.$$

We define the initial ideal of I with respect to w as the ideal generated by the initial forms of the elements of I , i.e.

$$\text{in}_w(I) = (\text{in}_w(f) \mid f \in I).$$

Note that the initial form does not need to be a monomial.

Example 2.6. Let I be the ideal generated by

$$f(x_1, x_2) = x_1^5 x_2 + x_1^4 x_2^4 + x_1^4 + x_1 x_2^5 + x_1 x_2^2 + x_2^6 + x_2$$

If $w = (1, 1)$ then $\text{in}_w(f) = x_1^4 x_2^4$, so $\text{in}_w(I)$ is a monomial ideal. If $w = (1, 2)$ then $\text{in}_w(f) = x_1^4 x_2^4 + x_1^2 x_2^5 + x_2^6$, so $\text{in}_w(I)$ is not a monomial ideal.

However, this is not a huge problem, because of the following result (which we will not prove).

Proposition 2.2. Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ be a monomial order. Then there exists $w \in \mathbb{N}^n$ such that $\text{in}_w(I) = \text{in}_{\succ}(I)$.

We say that w represents a $\succ$ for I .

2.4. The Gröbner fan.

Definition 2.9. We say $w, w' \in \mathbb{R}^n$ are **equivalent** with respect to some ideal I if

$$\text{in}_w(I) = \text{in}_{w'}(I)$$

Proposition 2.3. Given an ideal I , each equivalence class of weight vectors is a relatively open polyhedral cone (i.e. a polyhedral cone with its proper faces removed)

The proof of this statement relies on showing that

$$C[w] = \{w' \in \mathbb{R}^n \mid \text{in}_{w'}(g) = \text{in}_w(g) \text{ for all } g \in \mathcal{G}\}$$

where the left-hand side is the notation for the equivalence class of weights equivalent to w . On the right-hand side $\mathcal{G}$ is the reduced Gröbner bases with respect to $\succ_w$, the monomial ordering defined by w , where some other monomial ordering breaks ties.

This statement has a nice geometric interpretation

$$C[w] = \mathcal{N}_Q(F_w(Q))$$

where

$$Q = \text{Newt} \left(\prod_{g \in \mathcal{G}} g \right) = \sum_{g \in \mathcal{G}} \text{Newt}(g)$$

Definition 2.10. Let I be an ideal in $k[x_1, \dots, x_n]$. The **Gröbner fan** of I is the set of closed cones $C[\bar{w}]$ for all $w \in \mathbb{R}^n$.

Remark 2.1. It is not trivial to show that the Gröbner fan of I is actually a fan, but it is true.

Example 2.7. Consider the principal ideal $I = (xy + x + y) \subset k[x, y]$. Since I is principal, $\mathcal{G} = \{xy + x + y\}$ for any weight vector (up to scaling and choice of leading term), so $Q = \text{Newt}(xy + x + y) = \text{conv}\{(1, 1), (1, 0), (0, 1)\}$, a triangle. The three vertices correspond to the three monomials xy , x , and y .

For $w = (1, 2)$, we have $w \cdot (1, 1) = 3$, $w \cdot (1, 0) = 1$, $w \cdot (0, 1) = 2$, so $\text{in}_w(xy + x + y) = xy$ and $F_w(Q) = \{(1, 1)\}$. The equivalence class $C[w]$ is the relative interior of the normal cone of the vertex $(1, 1)$, which is $\{w' \in \mathbb{R}^2 \mid w'_1 > 0, w'_2 > 0\}$ – the open first quadrant. Every weight in this region makes xy the leading term.

For $w = (1, -1)$, the leading term is x (since $w \cdot (1, 0) = 1 > 0 = w \cdot (1, 1) > -1 = w \cdot (0, 1)$), and $C[w]$ is the relative interior of the normal cone of $(1, 0)$, namely $\{w' \mid w'_2 < 0, w'_1 > w'_2\}$.

For $w = (-1, 1)$, the leading term is y by symmetry, and $C[w] = \{w' \mid w'_1 < 0, w'_2 > w'_1\}$.

The three open cones, together with their boundary rays and the origin, partition $\mathbb{R}^2$ into the normal fan of the triangle Q , which is exactly the Gröbner fan of I .

On the wall $w_2 = 0$, $w_1 > 0$ (the boundary between the xy -cone and the x -cone), the initial form is $\text{in}_{(1,0)}(xy + x + y) = xy + x = x(y + 1)$, which is not a monomial – reflecting the fact that two terms of f are tied for maximal w -degree.

For a homogeneous ideal, the equivalent of the Newton polytope is the **state polytope**, which is a convex polytope whose normal fan is the Gröbner fan.